

**ED 6002: Optimization Methods in Engineering Design (2025-26)**

*Assignment 2 / Pranav D / ns26z167*

**Q1. Bracket Minima - Exhaustive Search**

**Objective Function:**

$$\text{Min. } f(x) = 2e^x - x^3 - 10x$$

**Design Space:**

$$x \in [0, 10]$$

**Initial point and increment:**

$$x_0 = 0 \triangle x = 2$$

**After Solving...**

- Terminated after 1st Iteration
- Minima lies between (0,4)

Q1) Min  $f(x) = 2e^x - x^3 - 10x$ .  
 $x \in [0, 10]$

$x_0 = 0$        $\Delta x = 2$

Using Exhaustive Search Method.

Iteration 1:

$x_0 = 0$        $x_1 = 2$        $x_2 = 4$

$f(x_0) = 2e^0 - 0^3 - 10(0) = 2$

$f(x_1) = 2e^2 - 2^3 - 10(2) = -13.222$

$f(x_2) = 2e^4 - 4^3 - 10(4) = 5.196$

$f(x_0) \geq f(x_1) \leq f(x_2)$

$\therefore$  Condition for Termination achieved.

So  $x^* \in (0, 4)$



## Q2. 2 Iterations of Golden Section Search

Q2) Two Iterations of Golden Search.

$$\text{Min } f(x) = 2e^x - x^3 - 10x$$

From the previous solution,

$$a=0 \quad b=4$$

Converting from  $x$  design space to  $w$  space.

$$w = \frac{x-0}{4-0}$$

$$\therefore x = 4w$$

$$f(w) = 2e^{4w} - (4w)^3 - 40w$$

$$a_w = 0 \quad b_w = 1 \quad L_w = 1$$

Iteration 1 :-

$$w_1 = 0 + 0.618(1)$$

$$w_2 = 1 - 0.618(1)$$

$$w_1 = 0.618$$

$$w_2 = 0.382$$

$$f(w_1) = -16.13$$

$$f(w_2) = -7.45$$

$$f(w_1) < f(w_2)$$

So eliminating the region from  $a$  to  $w_2$ .

$$\text{So } a_w = w_2 = 0.382 \quad b_w = 1 \quad L_w = 0.618$$



Iteration 2 :-

$$w_1 = 0.382 + 0.618(0.618)$$

$$w_2 = 1 - 0.618(0.618)$$

$$w_1 = 0.763$$

$$w_2 = 0.618$$

$$f(w_1) = -16.63$$

$$f(w_2) = -16.13$$

Again,  $f(w_1) < f(w_2)$

Eliminating the region from  $a$  to  $w_2$ .

Updating the bounds,

$$a_w = w_2 = 0.618, \quad b_w = 1, \quad L_w = 0.382$$

At the end of 2 iterations  $w^* \in (0.618, 1)$

Converting to design space  $x$ ,

$$\therefore x^* \in (2.472, 4)$$

Size of the bracketed interval = 1.528.

$$f(x_L) = f(2.472) = -16.134$$

$$f(x_U) = f(4) = 5.196$$

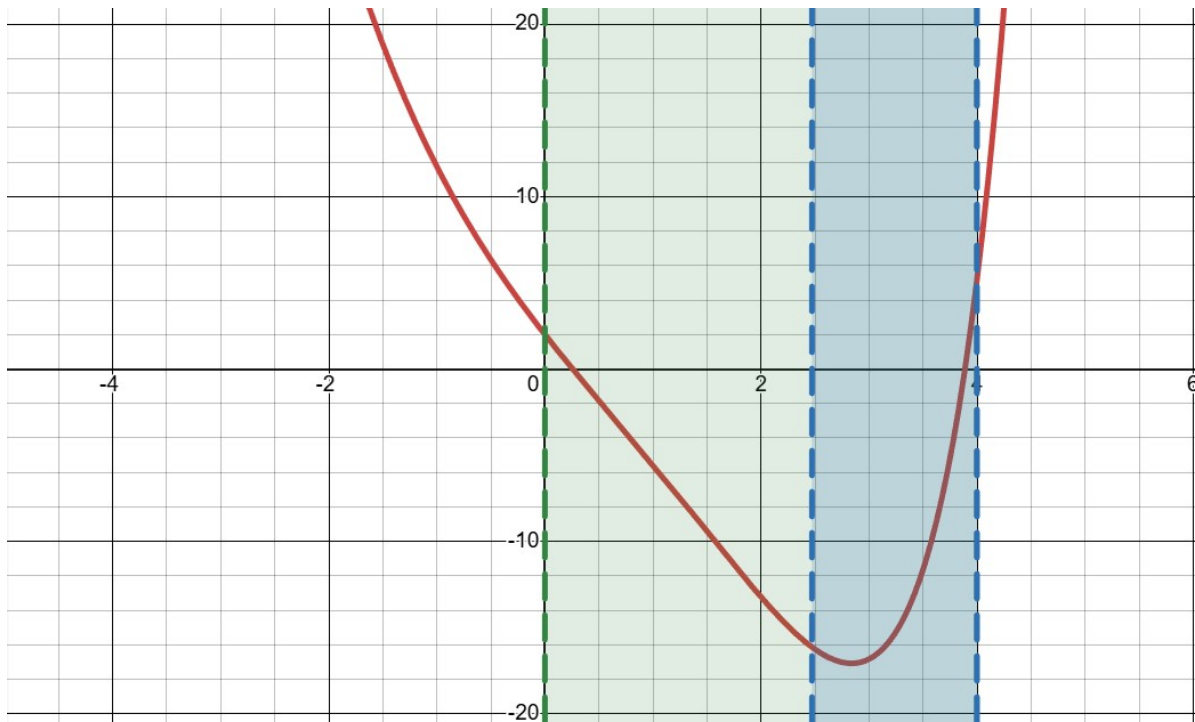
$$\Delta f = f(x_U) - f(x_L) = 5.196 + 16.134 = 21.33.$$

Difference between the function value.

at Interval bounds is 21.33.

### After Solving...

- Bounds shrunk to (2.472,4) (in blue)
- Size of the bracketed interval = 1.528
- Difference between the function value at bounds = 21.33





### Q3. Solving analytically

Q3) Given that,

$$f(x) = 2e^x - x^3 - 10x.$$

$$\frac{df}{dx} = \frac{d}{dx} (2e^x - x^3 - 10x) = 0.$$

$$2e^x - 3x^2 - 10 = 0.$$

Solving the equation gets,

$$x^* \approx 2.837.$$

Verify the nature using 2nd order optimality condition,

$$\frac{d^2f(x)}{dx^2} = \frac{d}{dx} (2e^x - 3x^2 - 10).$$

$$f''(x) = 2e^x - 6x.$$

$$f''(x^*) = 2e^{2.837} - 6(2.837) = 17.1$$

$$\therefore f''(x^*) > 0$$

Therefore  $x^*$  is a minimum point.

$x^*$  lies in between the brackets obtained from 2 iterations of Golden section search previously.



Additionally, We can assume that minima.  
lies in the middle of brackets obtained from.  
Golden section method,

$$\text{So } x_{gs}^* = \frac{2.472 + 4}{2} = 3.236$$

$$f(x_{gs}^*) = -15.38$$

$$\text{Meanwhile, } x_{true}^* = 2.837$$

$$f(x_{true}^*) = -17.07$$

So, difference between the results could be listed as,

$$\Delta = x_{true}^* - x_{gs}^* = -0.399$$

$$\Delta f = f(x_{true}^*) - f(x_{gs}^*) = -1.69$$

## Comments

- Results from the golden section search method and analytical approach were compared
- In Golden section search,
  - $x^* = 3.236$
  - $f(x^*) = -15.38$
- From analytical approach,
  - $x^* = 2.837$
  - $f(x^*) = -17.07$
- $x^*$  differs by -0.4, which resulted in -1.69 deviation from actual minimum value of the function

## Q4. Code for Exponential Search

```
function [a,b,k] = expsearch(f, x_init, d, t, max_iter)
    if nargin < 4
        max_iter = 100;
    end
    k = 0;
    if f(x_init - abs(d),t) >= f(x_init,t) && f(x_init,t) >= f(x_init + abs(d),t)
        d = abs(d);
    elseif f(x_init - abs(d),t) <= f(x_init,t) && f(x_init,t) <= f(x_init + abs(d),t)
        d = -abs(d);
    else
        disp("choose diff x_init or d")
        return
    end
    x_k = x_init;
    for i = 1:max_iter
        x_k1 = x_k + (2^k) * d;

        if f(x_k1,t) < f(x_k,t)
            k = k + 1;
            x_k = x_k1;
        else
            x_prev = x_k - (2^(k-1)) * d;
            a = min(x_prev, x_k1);
```

```

        b = max(x_prev, x_k1);
        return
    end
end
disp("max iterations")
a = min(x_prev, x_k1);
b = max(x_prev, x_k1);
end

```

### Code for Region Halving

```

function [a,b,l,k] = inthalving(f,a,b,t,e)
    if nargin < 5
        e = 0.1;
    end
    xm = (a+b)/2;
    l = b-a;
    k = 0;
    while abs(l) > e
        x1 = a + (l/4);
        x2 = b - (l/4);
        if f(x1,t) < f(xm,t)
            b = xm;
            xm = x1;
        elseif f(x2,t) < f(xm,t)
            a = xm;
            xm = x2;
        else
            a = x1;
            b = x2;
        end
        l = b-a;
        k = k+1;
    end
end

```

## Combine both & Benchmark

```
function y = sample_f(x, t)
    y = (2 * exp(x)) - (x.^3) - (10*x);
    % y = (x.^2) + (54./x);
end
```

```
sample_f = @sample_f;
[a_es,b_es,k_es] = expsearch(sample_f, 0, 2, 0, 100)
[a_ih,b_ih,l_ih,k_ih] = inthalving(sample_f, a_es, b_es, 0, 0.1)
```

```
a_es = 0           // Lower bound in Exp. Search
b_es = 6           // Upper bound in Exp. Search
k_es = 1           // Number of iters in Exp. Search
a_ih = 2.8125       // Lower bound in Interval halving
b_ih = 2.9062       // Upper bound in Interval halving
l_ih = 0.0938       // Final size of the Interval
k_ih = 6           // Number of iters in Interval halving
```

## Q5. Solving for Blackbox function

```
fn = @myfunction;

x_init = [2 0 2 5 0 0 6 0 0 10];
d_init = [5 1 5 10 1 1 1 1 1 1];

max_iter = 5000;
e = 0.1;

Case = (1:10).';
x_init = x_init(:);
d = d_init(:);

a_es = nan(10,1);
b_es = nan(10,1);
```



```

k_es    = nan(10,1);

a_ih    = nan(10,1);
b_ih    = nan(10,1);
l_ih    = nan(10,1);
k_ih    = nan(10,1);

optima = nan(10,1);
fn_val = nan(10,1);

for c = 1:10
    [a_es(c), b_es(c), k_es(c)] = expsearch(fn, x_init(c), d(c), c, max_iter);
    [a_ih(c), b_ih(c), l_ih(c), k_ih(c)] = inthalving(fn, a_es(c), b_es(c), c, e);

    optima(c) = (a_ih(c) + b_ih(c)) / 2;
    fn_val(c) = fn(optima(c), c);
end

Results = table(Case, x_init, d, a_es, b_es, k_es, a_ih, b_ih, l_ih, k_ih, optima, fn_val);
disp(Results)

```

#### Q5. Comparing the solutions

| Case | x_init | d  | a_es        | b_es | k_es | a_ih    | b_ih    | l_ih     |
|------|--------|----|-------------|------|------|---------|---------|----------|
| ---- | -----  | -- | -----       | ---- | ---- | -----   | -----   | -----    |
| 1    | 2      | 5  | -13         | 2    | 1    | -10.334 | -10.275 | 0.058594 |
| 2    | 0      | 1  | 0           | 3    | 1    | 0.42188 | 0.51562 | 0.09375  |
| 3    | 2      | 5  | -13         | 2    | 1    | -5.5293 | -5.4707 | 0.058594 |
| 4    | 5      | 10 | 5           | 35   | 1    | 26.709  | 26.768  | 0.058594 |
| 5    | 0      | 1  | 1           | 7    | 2    | 2.9688  | 3.0625  | 0.09375  |
| 6    | 0      | 1  | 0           | 3    | 1    | 0.9375  | 1.0312  | 0.09375  |
| 7    | 6      | 1  | 5           | 6.5  | 0    | 5.0469  | 5.1406  | 0.09375  |
| 8    | 0      | 1  | 0           | 3    | 1    | 2.9062  | 3       | 0.09375  |
| 9    | 0      | 1  | 1           | 7    | 2    | 2.7812  | 2.875   | 0.09375  |
| 10   | 10     | 1  | 4.4942e+307 | Inf  | 1023 | Inf     | NaN     | NaN      |

## Comments

- Case 7 may contain a random fn
- Case 10 is just a straight line (100-2x)

```
fn = @myfunction;

x10 = (-3:3)';
vals10 = arrayfun(@(t) fn(t,10), x10);
Sol10 = [x10 vals10]

x7=zeros(5,1);
vals7 = arrayfun(@(t) fn(t,7), x7);
Sol7 = [x7 vals7]
```

```
Sol10 =
    -3    106
    -2    104
    -1    102
     0    100
     1     98
     2     96
     3     94
Sol7 =
     0     0
     0     1
     0     1
     0     1
     0     0
```